HEALTHCARE BUSINESS ANALYSIS

Multi-Hospital Patient Performance Report

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Project Date: December 2025

Tool Used: Microsoft Excel

Dataset: 55,500 Patient Records| 2019-2024

1. Project Overview

This report presents findings from a comprehensive data analytics engagement for a large U.S. healthcare network. The objective was to analyse patient performance data across multiple hospitals and generate actionable insights for executive decision-making.

The entire analysis was performed in Microsoft Excel using Pivot Tables, Pivot Charts, and an interactive Dashboard with Slicers. The dataset comprised 55,500 patient records spanning six years (2019-2024).

Key Deliverables

- Cleaned and structured the dataset with derived columns
- 10 Pivot Tables and Pivot Charts covering all six business questions
- Fully interactive Excel Dashboard with Gender and Age Group slicers
- Executive summary with key insights and strategic recommendations

2. Business Problem Statement

The executive team identified six core analytical questions requiring data-driven answers:

#	Business Question	Category
1	How are patient numbers changing over time, by age group, gender, admission type, and hospital?	Patient Volume Trends
2	What is driving the average treatment cost and hospital stay duration?	Cost & Stay Analysis
3	Which hospitals have the highest or lowest patient volumes, and why?	Hospital Performance
4	What medications are commonly administered, and what are the test result distributions?	Medication & Test Patterns
5	How does patient volume break down by insurance provider?	Insurance Impact
6	Are there seasonal patterns in admissions affecting staffing and resources?	Seasonality Analysis

3. Dataset Description

The raw dataset contained 55,500 rows and 15 columns, representing anonymised patient records from multiple U.S. hospitals from 2019 to 2024.

Column Name	Data Type	Description
Name	Text	Patient full name (anonymized)
Age	Numeric	Patient age in years
Gender	Text	Male / Female
Blood Type	Text	Patient blood group (A+, A-, B+, etc.)
Medical Condition	Text	Primary diagnosis
Date of Admission	Date	Hospital admission date
Doctor	Text	Attending physician name
Hospital	Text	Hospital name
Insurance Provider	Text	Insurance company name
Billing Amount	Numeric	Total treatment cost in USD
Room Number	Numeric	Assigned room number
Admission Type	Text	Emergency / Urgent / Elective
Discharge Date	Date	Patient discharge date
Medication	Text	Primary medication administered
Test Results	Text	Normal / Abnormal / Inconclusive

4. Data Cleaning Process

All data cleaning was performed in Microsoft Excel before creating Pivot Tables.

4.1 Text Standardisation

- Patient names, hospital names, and doctor names had inconsistent capitalisation. PROPER() function was applied to standardise all text fields.

4.2 Date Validation

- Date of Admission and Discharge Date columns were verified and formatted consistently in YYYY-MM-DD format.

4.3 Derived Columns Created

- Length of Stay (Days) - calculated as Discharge Date minus Date of Admission using DATEDIF()
- Year - extracted from Date of Admission using YEAR() for trend analysis
- Month - extracted using MONTH() for seasonality analysis
- Age Group - binned into 7 categories (0-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71+) using nested IF() formulas

4.4 Duplicate and Null Check

- Excel's Remove Duplicates feature confirmed no duplicate records existed.
- No blank values were found in critical columns: Gender, Admission Type, Insurance Provider, or Medical Condition.

5. Dashboard and KPI Summary

An interactive Excel Dashboard was created on a dedicated sheet, displaying all key metrics and charts in a single view with KPI cards at the top and 9 visualisations below.

5.1 Key Performance Indicators (KPIs)

KPI Metric	Value	Significance
Total Patients	55,500	Complete patient volume across all hospitals and years
Total Treatment Cost	$1,41,74,32,043.4	Cumulative billing across the entire network
Avg Treatment Cost per Patient	$25,539.32	Benchmark for cost efficiency analysis
Avg Length of Stay	16 Days	Operational efficiency and bed utilization metric

5.2 Interactive Slicers

Two slicers provide dynamic filtering across all charts and KPI cards:

- Gender Slicer - filter all visualisations by Male or Female patients.
- Age Group Slicer - filter by age bracket from 0-20 through 71+.

6. Analysis and Findings

6.1 Yearly Admission Trend

Patient admissions grew significantly from 2019 to 2020, then stabilised at around 11,000 per year through 2023. The 2024 figure reflects partial-year data only.

Year	Patient Admissions	Trend
2019	7,387	Base Year
2020	10,931	+48% growth
2021	11,017	Peak Year
2022	11,285	Stable
2023	11,026	Stable
2024	3,854	Partial Year

6.2 Gender Distribution

Female: 27,726 patients (49.95%)

Male: 27,774 patients (50.05%)

The near-perfect gender balance validates the representativeness of the dataset.

6.3 Medical Conditions

All six medical conditions are distributed with near-equal frequency, indicating the network serves a broad clinical spectrum.

Medical Condition	Patient Count	% Share
Arthritis	9,308	16.77%
Diabetes	9,304	16.76%
Hypertension	9,245	16.66%
Obesity	9,231	16.63%
Cancer	9,227	16.62%
Asthma	9,185	16.55%

6.4 Admission Type Breakdown

Emergency: 18,269 patients (32.9%)

Urgent: 18,576 patients (33.5%)

Elective: 18,655 patients (33.6%)

The near-equal distribution indicates a well-balanced intake system. Combined Emergency and Urgent cases (66.4%) highlight the need for strong emergency readiness.

6.5 Insurance Provider Analysis

Insurance Provider	Patient Count	% Share
Cigna	11,249	20.27%
Medicare	11,154	20.10%
UnitedHealthcare	11,125	20.05%
Blue Cross	11,059	19.93%
Aetna	10,913	19.66%

All five providers have near-identical volumes, ensuring a diversified and balanced payer mix for the network.

6.6 Age Group Distribution

Age Group	Patient Count
0-20	2,443
21-30	8,056
31-40	8,125
41-50	8,209
51-60	8,297
61-70	8,228
71+	12,142

The 71+ age group is the single largest patient segment, underscoring elder care as a critical operational priority.

6.7 Medication and Test Results

All five medications are prescribed at near-uniform frequency: Lipitor (11,140), Ibuprofen (11,127), Aspirin (11,094), Paracetamol (11,071), Penicillin (11,068).

Test result breakdown: Abnormal 18,627 (33.6%), Normal 18,517 (33.4%), Inconclusive 18,356 (33.1%). The high Abnormal and Inconclusive rates warrant structured clinical follow-up protocols.

6.8 Hospital and Doctor Performance

The top 5 hospitals by patient volume are all part of the Smith network group. Top performing doctors by patient volume include Michael Smith (27 patients), John Smith (22), and Robert Smith (22).

7. Strategic Recommendations

7.1 Elder Care Investment

- With 71+ representing the largest segment (12,142 patients), prioritise geriatric wards, specialist hiring, and elder-focused care pathways.

7.2 Emergency Readiness

- 66.4% of admissions are Emergency or Urgent. Ensure 24/7 staffing, adequate ICU capacity, and emergency supply chain robustness.

7.3 Insurance Payer Strategy

- Maintain strong contract negotiations with all 5 insurance providers to protect the current balanced reimbursement structure.
- Medicare's 20.1% share requires careful coding accuracy and compliance to maximise reimbursements.

7.4 Chronic Disease Management

- Arthritis, Diabetes, and Hypertension collectively account for approximately 50% of all cases. Dedicated chronic disease management and prevention programs can reduce readmissions and costs.

7.5 Test Result Follow-up Protocol

- 33.6% Abnormal and 33.1% Inconclusive results demand an automated patient recall system to ensure timely clinical follow-up and improved outcomes.

7.6 Cost Optimization

- With Avg Treatment Cost at $25,539, review high-cost admission types and medical conditions for cost reduction opportunities through protocol standardisation.

8. Tools and Methodology

Tool / Feature	Purpose
Microsoft Excel - Data Sheet	Raw data storage, cleaning, and derived column creation
Excel Pivot Tables	Aggregation and cross-tabulation of patient data
Excel Pivot Charts	Visual representation of all pivot analyses
Excel Slicers	Interactive dashboard filtering by Gender and Age Group
DATEDIF() / YEAR() / MONTH()	Date calculations and time-based feature engineering
IF() / COUNTIFS() / SUMIFS()	Conditional calculations and KPI computation
AVERAGEIFS()	Conditional averages for billing and stay duration
Remove Duplicates / Data Validation	Data quality and integrity checks

9. Conclusion

This Excel-based healthcare analytics project successfully transformed 55,500 raw patient records into a structured, interactive, and insight-rich dashboard. All six business questions posed by the executive team have been comprehensively addressed through data-driven analysis.

The analysis reveals a well-distributed patient population across genders, age groups, medical conditions, and insurance providers. The key strategic priorities are elder care investment, emergency readiness, chronic disease management, and a robust test result follow-up system.

The interactive Excel dashboard provides the executive team with a self-service analytical tool that can be refreshed with new data to ensure ongoing visibility into the network's performance.